

Tide retrospective: *2D partition function asymptotic (Z2)*

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Abstract

Completes today's V2/Z1/Z2 triple — moment, numerator, partition, all packaged as exact reformulation + rescaled `Tendsto` + `IsEquivalent` for the 2D quartic-sextic potential. Z2 reduces to Z1 specialised to $(j, k) = (0, 0)$: the seabed already supplies the exact closed form for the 2D partition function, so this Tide is purely the asymptotic-packaging step. 125 lines, single-attempt clean build, no roadblocks, no GPT consult (third consecutive mirror-shaped tide today). Zero `sorry`, zero `axiom`, zero `native_decide`.

1 Setting

V2 packaged T1's exact moment closed form; Z1 packaged the unnormalised numerator's exact closed form. The partition function is the natural third member of the triple: $Z_{2D}(t) = \iint e^{-tL(x,y)} dx dy$ for $L(x,y) = x^4/24 + y^6/720$. Its exact closed form was already in the seabed, named `partitionFunction_quarticSextic_eq`; Z2 just adds the asymptotic packaging on top.

The seabed dig surfaced a stronger result than expected: the exact 2D partition closed form is right there in the same file as A2's factor theorem.¹ So Z2 is literally Z1 at $(j, k) = (0, 0)$ with the seabed-provided exact identity replacing Z1's separability-composition step. Even cleaner.

2 The thing formalised

Three theorems in `Laplace.TwoD`:

1. **Exact reformulation.** For $t > 0$, $Z_{2D}(t) = C'' \cdot t^{-1/4-1/6}$ with $C'' = \frac{1}{6} \cdot 24^{1/4} \cdot 720^{1/6} \cdot \Gamma(1/4) \cdot \Gamma(1/6)$.
2. **Rescaled Tendsto.** As $t \rightarrow \infty$, $t^{1/4+1/6} \cdot Z_{2D}(t) \rightarrow C''$.
3. **IsEquivalent.** At `atTop`, $Z_{2D}(t) \sim C'' \cdot t^{-1/4-1/6}$.

Named constant: `quarticSexticPartitionConst`. File: sibling to V2's and Z1's asymptotic files, in the same `Laplace/TwoD/` folder.²

¹In `QuarticSextic.lean`, named `partitionFunction_quarticSextic_eq`, sibling to `partitionFunction_quarticSextic_factor` and T1's moment factor theorem.

²Specifically `QuarticSexticPartitionAsymptotic.lean`, alongside `QuarticSexticMomentAsymptotic.lean` (V2) and `QuarticSexticNumeratorAsymptotic.lean` (Z1).

3 Proof strategy

For the exact reformulation: one `rw` of the seabed’s existing 2D-partition closed form (giving $(1/2)(24/t)^{1/4}\Gamma(1/4) \cdot (1/3)(720/t)^{1/6}\Gamma(1/6)$), then the same `Real.rpow` algebra V2 and Z1 used: `div_rpow` splits $(24/t)^{1/4}$, `rpow_add` combines $t^{-1/4} \cdot t^{-1/6} = t^{-1/4-1/6}$, `rpow_neg` bridges the negative exponents, and `ring` closes. The two asymptotic forms follow by the same eventual-equality idiom.

4 Roadblocks and resolutions

None. Z2 is the simplest of today’s three mirror-shaped tides: a single seabed identity rewrite plus the standard `rpow` collapse. No `field.simp` pitfall this time (lesson from V2 carried over). `lake` build green on first compile.

5 What was Mathlib, what was new

From Mathlib. The same set of API V2/Z1 used: the `Real.rpow` algebra suite (`div_rpow`, `rpow_add`, `rpow_neg`), the `Tendsto-to-constant` idiom, and `IsEquivalent`’s `refl+congr_left` pair.

From the seabed. One lemma: the existing exact 2D partition closed form.³ Nothing else.

New here. Three theorems plus the named constant `quarticSexticPartitionConst`. The architectural value is in completing the V2/Z1/Z2 triple — moment / numerator / partition all now have parallel asymptotic packagings, in three sibling files.

6 Lessons

Three is the right cadence for mirror-shaped tides. V2 established the packaging shape against a freshly-deliberated GPT consult; Z1 mirrored it with one structural change (the seabed dig that found the existing factor lemma); Z2 mirrored both at minimal cost. The pattern is reproducible: once a packaging shape has been deliberated for one object, applying it to closely related objects is cheap and skip-GPT-safe. Worth recording as a tide-template recipe.

The 2D quartic-sextic story has a natural “triple” shape. Moment, numerator, partition are the three objects to which the asymptotic-packaging recipe applies; the partition is the “unit” object ($j = k = 0$ in the numerator family), so Z2 specialises Z1. Future programmes on other separable 2D potentials should be able to ship the same three-theorem-times-three structure with minimal additional deliberation.

7 Follow-ups

- *Generic- k 1D and 2D analogues.* The whole moment / numerator / partition triple should extend to pure $x^{2k}/(2k)!$ potentials in 1D (using N1’s J-function machinery) and to the separable 2D combinations of any pair of such 1D potentials. Out of scope for a single Tide; this would be a programme that the V2/Z1/Z2 triple has effectively prototyped.

³`partitionFunction_quarticSextic.eq` in `Laplace/TwoD/QuarticSextic.lean`.

- *Promote the three named constants to a single shared definition.* `quarticSexticMomentConst`, `quarticSexticNumeratorConst`, and `quarticSexticPartitionConst` share structure; a single parametrised definition might consolidate them. Currently each is used once; defer until a fourth consumer appears.
- *\leanref-tag the primer / grammar paper* with the full V2/Z1/Z2 triple. Pairs with the existing paper-tagging backlog.

Acknowledgements

No GPT consult was used for Z2 (third mirror-shaped tide of the day). The deliberation work inherited from V2's consult, which set the three-theorem packaging shape and the `tendsto_congr`' idiom, and applied unchanged here. No numerical check was needed beyond inheriting Z1's $(j, k) = (0, 0)$ row, which verified the closed form to relative error $\sim 10^{-16}$.